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Techno-Economic Assessment of Ammonia-Fueled Combined Cycle Gas Turbines for Power Generation in Saudi Arabia

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Abstract

Ammonia is rapidly emerging as a leading candidate among synthetic fuels for low-carbon power generation, offering carbon-free combustion and favorable energy density. It can be produced from renewable sources such as solar and wind through green hydrogen pathways and is easier to store and transport than hydrogen itself. With increasing global interest in decarbonizing baseload electricity systems, ammonia presents a scalable, energy-dense, and technically viable fuel alternative to natural gas. This study assesses its feasibility in combined cycle gas turbine (CCGT) applications tailored for conditions in the Kingdom of Saudi Arabia.

A comprehensive techno-economic assessment was conducted using Aspen HYSYS to simulate a 2×2×1 H-class CCGT system retrofitted to combust ammonia. The model captured combustion dynamics, turbine performance, steam cycle integration, and heat rate variations due to ambient temperature effects. Economic inputscapital expenditure (CAPEX), operational expenditure (OPEX), and fuel delivery costs—were derived from NETL, PJM, and EIA data and escalated to 2022 USD using CEPCI multipliers. Scenarios were evaluated for both greenfield installations and brownfield retrofits of natural gas systems. The financial evaluation used a discounted cash flow (DCF) model to compute levelized cost of electricity (LCOE).

Results indicate that while ammonia incurs higher fuel and NO_x abatement costs than natural gas, its LCOE can approach competitiveness under specific conditions—particularly in retrofit cases and with supportive carbon pricing. Ammonia-fired CCGT systems exhibit resilience under high-temperature ambient conditions typical of Saudi Arabia, with only marginal heat rate penalties. The study concludes that ammonia, when supported by infrastructure investment and carbon policy frameworks, can be a cornerstone fuel in transitioning to a low-carbon utility power sector.

Introduction

The global push for decarbonization has intensified the search for scalable, carbon-free energy carriers that can complement intermittent renewables. Among the candidates, ammonia (NH₃) stands out due to its high energy density, carbon-free combustion, and established infrastructure for storage and transportation

[Valera-Medina et al., 2018]. As a hydrogen-rich molecule, ammonia offers a practical means to store and utilize hydrogen energy, especially in power systems where electrification alone cannot meet reliability and baseload requirements [IEA, 2021]. These attributes make ammonia an attractive fuel for combined cycle gas turbine (CCGT) systems, which are widely used for dispatchable electricity generation worldwide.

Ammonia's direct combustion, however, introduces specific engineering challenges. Its low flame speed and high ignition temperature can lead to combustion instability, while high nitrogen content results in elevated NO_x emissions. Nevertheless, recent advancements in combustor design, staged fuel injection, and integration of selective catalytic reduction (SCR) systems have made ammonia combustion increasingly viable [Kobayashi et al., 2019; Cheng et al., 2022]. Experimental studies on micro gas turbines and ammonia co-firing projects—such as those conducted by IHI Corporation and Mitsubishi Heavy Industries—demonstrate that ammonia can be reliably burned at various load conditions [Okafor et al., 2021; IHI, 2020]. These findings align with simulation work using ASPEN HYSYS and similar platforms, validating thermodynamic performance models and off-design behavior [Liu & Karimi, 2018].

Multiple techno-economic analyses have explored the financial viability of ammonia-fired generation. Cesaro et al. (2021) projected the LCOE for green ammonia-fueled plants and concluded that under favorable policy and fuel price scenarios, ammonia could be competitive with natural gas by the 2030s. Zhang et al. (2021) performed a comparative assessment of hydrogen and ammonia CCGTs, finding that ammonia systems had lower storage and handling costs, particularly for utility-scale deployment. More recently, Wang et al. (2023) assessed retrofit strategies and emphasized that CCGT plants with dual-fuel burners and moderate SCR upgrades could switch to ammonia with less than 10% CAPEX increase. However, these studies often focus on temperate climates or assume Western capital cost structures, which differ significantly from Saudi Arabia's labor and fuel economics.

In the Middle East context, Saudi Arabia is uniquely positioned to scale ammonia-based power generation. The Kingdom's Vision 2030 prioritizes diversification of energy sources and massive investment in green hydrogen and ammonia infrastructure, such as the NEOM Green Hydrogen Project, which aims to produce 1.2 million tons of green ammonia annually. Domestic electricity demand is also projected to rise significantly due to industrial growth and climate-driven air conditioning loads [Saudi Energy Efficiency Center, 2022]. These factors necessitate reliable, low-carbon baseload generation, and ammonia-fired CCGTs could play a strategic role—especially if they can leverage existing NG infrastructure through retrofits.

This study addresses a key knowledge gap by performing a techno-economic assessment of ammonia-fired CCGTs specifically under Saudi Arabian conditions. Using a validated ASPEN HYSYS model of a 2×2×1 H-Class CCGT (400 MW), the study evaluates performance losses under ammonia firing, CAPEX and OPEX impacts of fuel switching, and the resulting LCOE across greenfield and retrofit cases. The findings offer practical insights for energy policymakers and project developers in Saudi Arabia and the Gulf Cooperation Council (GCC), showing under what conditions ammonia can deliver cost-effective, carbon-free electricity.

Recent reports from the International Energy Agency emphasize the strategic role of low-carbon hydrogen and ammonia in decarbonizing global power systems, highlighting their potential to complement intermittent renewables while maintaining system reliability [IEA, 2023]. Furthermore, experimental studies continue to demonstrate advancements in ammonia combustion stability and emissions control, reinforcing ammonia's viability for large-scale power generation [Huang et al., 2023].

Background & Motivation

The global energy transition is increasingly shaped by the need to decarbonize while maintaining reliable and flexible power supply. While renewables such as wind and solar dominate new generation capacity, their variability necessitates backup sources that are both clean and dispatchable. Ammonia, a zero-carbon

energy carrier that can be combusted directly or used as a hydrogen vector, is gaining international momentum as a promising solution [Valera-Medina et al., 2018]. Its compatibility with existing thermal power infrastructure, ease of transport and storage, and potential for large-scale production via green hydrogen pathways make it especially suitable for countries aiming to achieve net-zero targets while maintaining energy security.

Mass flows in the ammonia supply chain: from fossil fuel feedstocks to nitrogen fertilisers and industrial products

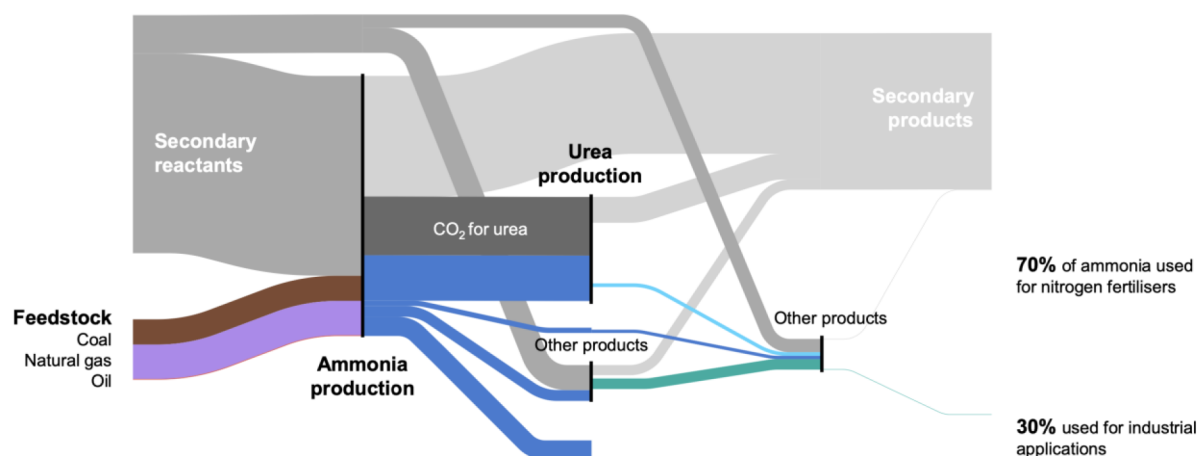


Figure 1—Global ammonia supply chain mass flows, illustrating the conversion of fossil fuel feedstocks into nitrogen fertilizers and industrial products. Adapted from IEA (2021), *Ammonia Technology Roadmap*.

Ammonia's role in power systems has evolved rapidly in recent years. Once considered primarily as an industrial chemical or fertilizer feedstock, it is now being tested in pilot-scale combustion systems worldwide. Projects such as IHI's 20 MW co-firing demonstration and Japan's national roadmap for ammonia power reflect growing institutional confidence in ammonia's technical feasibility [IHI Corp., 2020; METI, 2021]. Several academic studies have also modeled ammonia-fueled combined cycle gas turbines (CCGTs), showing that efficiency losses are manageable and that NO_x emissions can be mitigated using existing technologies like staged combustion and selective catalytic reduction (SCR) [Kobayashi et al., 2019; Cheng et al., 2022].

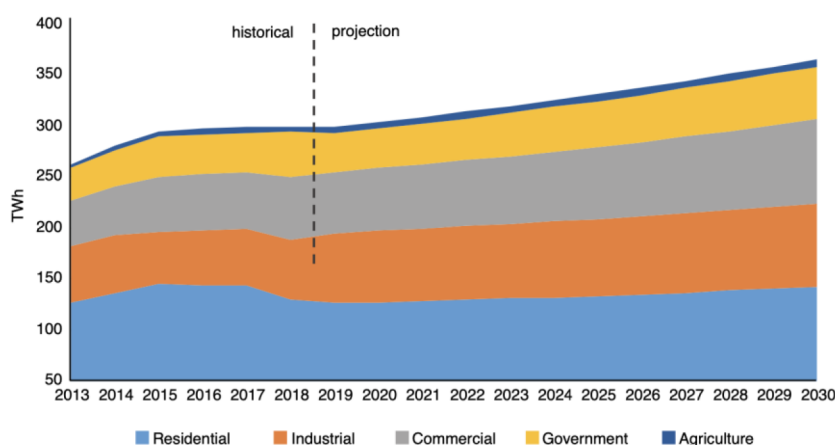
What distinguishes ammonia from other synthetic fuels is its production flexibility and infrastructure-readiness. Green ammonia can be synthesized using nitrogen and renewable hydrogen, while blue ammonia leverages natural gas with carbon capture. Moreover, unlike hydrogen, ammonia can be liquefied at −33°C under atmospheric pressure or stored at 10 bar at ambient temperature—conditions much more practical for existing pipelines and tank farms. This makes ammonia an ideal candidate for large-scale seasonal energy storage and cross-border trade, as acknowledged by recent IEA reports and European hydrogen strategy documents [IEA, 2021; European Commission, 2020].

Table 1—Comparison of Hydrogen Carriers – Energy Density & Storage Cost

Hydrogen Carrier	Gravimetric H ₂ Content	Volumetric Energy Density	Typical Storage Conditions	Storage & Transport Cost (USD/kg H ₂)	Key Pros	Key Cons
Ammonia (NH ₃)	~17.8 wt%	~12.7 MJ/L	-33°C @ 1 bar (liquid)	\$1.9–2.2	High density, mature infrastructure, low cost	Toxic, requires cracking
Liquid Hydrogen (LH ₂)	100 wt%	~8.5–10 MJ/L	-253°C (cryogenic)	\$2.0–3.0	High purity H ₂ , fast fueling	Expensive liquefaction, boil-off losses
LOHC (e.g., MCH)	~6 wt%	~5.5 MJ/L	Ambient temp, moderate pressure	\$2.0–2.5	Safe, diesel-like handling, reversible	Low H ₂ density, needs dehydrogenation
Methane (CH ₄ / Synthetic)	~25 wt%	~22 MJ/L (liquid)	Ambient (compressed or liquefied)	\$2.2–2.8	Existing infrastructure, high density	Not zero-carbon, CO ₂ emissions unless CCUS used

Saudi Arabia, in particular, offers a unique context for ammonia deployment. The Kingdom is both a high-potential producer and a high-demand consumer of dispatchable power. Mega-projects such as NEOM aim to produce over a million tons of green ammonia annually, leveraging abundant solar and wind resources to power electrolyzers. At the same time, national electricity demand is projected to grow by over 30% by 2030, driven by industrial growth, desalination, and year-round cooling loads. While gas-fired CCGTs currently dominate the generation mix, retrofitting them for ammonia could decarbonize a large portion of the grid with minimal new infrastructure [SEEC, 2022].

In addition to established benefits, recent research has underscored ammonia's competitive edge as a hydrogen carrier for dispatchable power, particularly when integrated with existing thermal infrastructure [Dincer et al., 2022; Giampaolo et al., 2022]. These studies further validate ammonia's practicality for countries such as Saudi Arabia, where grid reliability, hydrogen economy development, and infrastructure readiness are national priorities.

**Figure 2—Saudi Arabia Power Generation Mix and Forecasted Electricity Demand Growth**

This study responds directly to that gap. It conducts a detailed techno-economic assessment of ammonia-fired CCGTs using localized assumptions for CAPEX, OPEX, and ambient temperatures. It evaluates both greenfield and retrofit pathways and simulates performance using Aspen HYSYS, incorporating combustion dynamics, turbine output variation, and heat rate penalties under high-temperature conditions. Furthermore, the study quantifies the levelized cost of electricity (LCOE) for ammonia relative to natural gas and identifies policy interventions—such as carbon pricing and CAPEX subsidies—that could make ammonia

competitive. In doing so, it offers actionable insights for Saudi energy planners, utility operators, and clean fuel investors.

Methodology

To assess the viability of ammonia as a fuel in utility-scale power generation, this study conducted an integrated techno-economic analysis based on a 2×2×1 combined cycle gas turbine (CCGT) configuration using H-class gas turbines. This setup represents a modern and efficient architecture for large-scale power generation and was selected based on its widespread use in the Middle East and inclusion in benchmarking studies by the U.S. Energy Information Administration (EIA, 2020) and the National Energy Technology Laboratory (NETL, 2019). The specific model selected simulates a 400 MW net output CCGT plant, with two gas turbines, two heat recovery steam generators (HRSGs), and one steam turbine generator. Performance simulations were carried out using Aspen HYSYS, a widely used platform for process modeling and energy systems analysis.

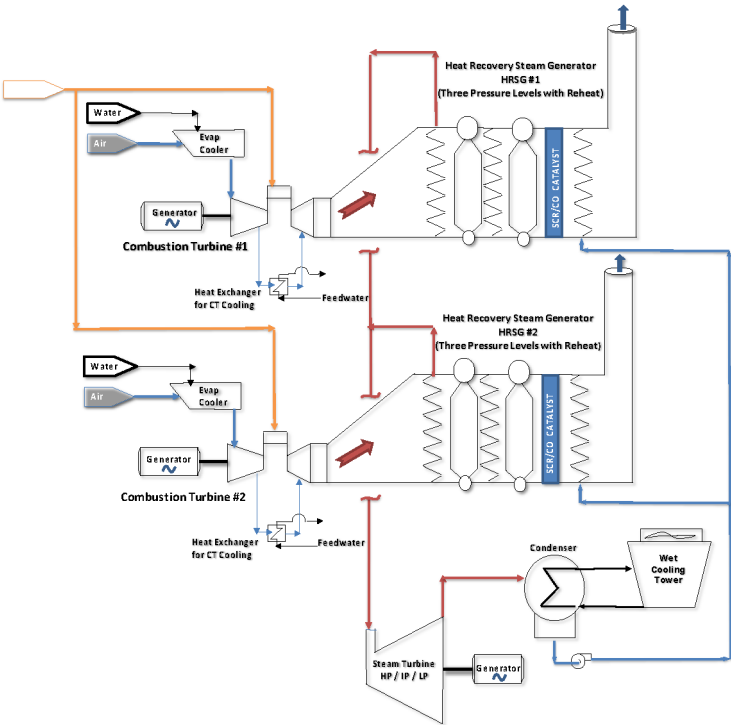
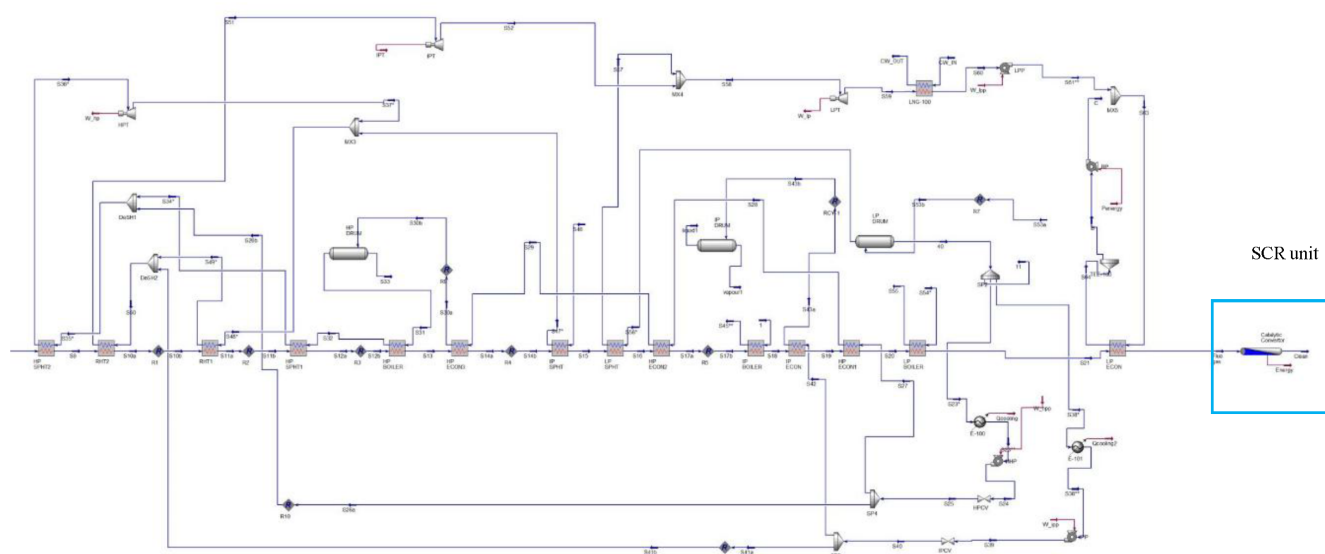


Table 2—Gas Turbine Operation Parameters

Category		Parameter	Value
Ambient Condition	Pressure		101.3 kPa
	Temperature		32.0 °C
Fuel Condition	Pressure		30.0 bar
	Temperature		10.0 °C
Gas Turbine	Inlet Air Flow		635.0 kg/s
	Compressor Pressure Ratio		15.4
	Fuel Flow		14.74 kg/s
	Combustor Efficiency		99.5%
	Combustor Exit Temperature		1405.0 °C
	Turbine Inlet Temperature		1328.0 °C
	Turbine Exhaust Temperature		615.0 °C

The diagram illustrates a gas turbine engine system with an ammonia fuel system. The process flow is as follows:

- Air Inlet:** Air enters through an AFT (Air Flow Transducer) and is then compressed by an Air Compressor.
- Ammonia Fuel System:** Pure ammonia (liquid) is supplied to an Ammonia Compressor, which is highlighted with a blue box. The output of the ammonia compressor is pure ammonia (pressurized), which is then fed into the Gibbs Combustor.
- Compressor Section:** The compressed air (S1) and pressurized ammonia (S3) enter the Gibbs Combustor. The output of the combustor is a gas stream (S4) that enters Mixer 1.
- Gas Turbine:** The gas stream (S4) enters the Gas turbine, which produces power (Wt) and exhausts a gas stream (S5).
- Separator and Cooling:** The gas stream (S5) enters Separator 1. The output of the separator is a gas stream (S2) that enters Mixer 2. The output of Mixer 2 is a gas stream (S7) that enters the DUCT. The output of the DUCT is a gas stream (S8) that is exhausted.
- Stator Cooling:** The gas stream (S2) is also used for stator cooling (S3).
- Rotor Cooling:** The gas stream (S2) is also used for rotor cooling (S7).
- Energy Stream:** An energy stream is fed into the Gibbs Combustor.
- Liquid Output:** The Gibbs Combustor also produces a liquid output.

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(CEPCI). The cost components were broken down into bare erected costs (BEC), engineering and procurement fees (EPC), contingencies, owner's costs, and total overnight cost (TOC). CEPCI subindices were used to individually escalate equipment, material, and labor costs (e.g., equipment multiplier = $1015.3 / 751.2 = 1.35$). Retrofit costs were modeled by removing redundant CAPEX elements and adding costs for fuel system upgrades and NO_x abatement.

Table 3—(EIA, NETL, PJM comparison, escalated to 2022 USD)

	EIA report		NETL reports				PJM report			
<i>Report Year</i>	2020		2019		2013				2018	
<i>Model</i>	H Class		F Class		F Class		H Class		H Class	
<i>Net Output</i>	1,083MW		726MW		631.6MW		821.1MW		1,138MW	
<i>CEPCI Year</i>	2020	2022	2018	2022	2011	2022	2011	2022	2017	2022
<i>Total BEC</i>	793,000	1,045,976	407,554	516,598	354,107	506,203	421,747	602,794	653,500	873,270
<i>EPC Fees</i>	79,300	104,597	81,510	103,319	28,577	101,240	34,042	120,559	65,300	87,330
<i>Contingency</i>	87,230	124,402	77,906	95,697	45,466	45,466	53,743	53,743	71,900	96,060
<i>TPC</i>	959,530	1,274,976	566,971	715,615	428,151	652,910	509,532	777,096	790,700	1,056,660
<i>Owner's Cost</i>	78,409	93,446	143,408	150,590	143,114	146,804	170,262	174,653	136,900	189,700
<i>TOC</i>	1,037,939	1,368,422	691,642	866,206	524,419	799,714	624,051	951,749	927,600	1,246,360
<i>Unit TOC</i>	\$958/kW	\$1264/kW	\$952/kW	\$1193/kW	\$830/kW	\$1266/kW	\$760/kW	\$1159/kW	\$885/kW	\$1214/kW

Escalation of Capital Costs Using CEPCI: Historical Trends and 2022 Baseline Adjustments

To normalize capital cost estimates across different years and sources, this study applied the Chemical Engineering Plant Cost Index (CEPCI) as a standard escalation metric. A consolidated CEPCI data summary was compiled from NETL (2013, 2019), PJM (2018), and EIA (2020) benchmarks, with February 2022 CEPCI values used as the baseline for adjusting historical estimates. As shown in the table, the overall CEPCI index rose from an average of 585.7 in 2011 to 801.3 in 2022, reflecting significant cost escalation in capital-intensive components.

Darker green rows in the table represent the four primary CEPCI subindexes: Equipment, Construction Labor, Buildings, and Engineering Supervision. Lighter green rows denote the detailed component indexes within the Equipment category, such as Heat Exchangers & Tanks, Process Machinery, Pipe & Fittings, and Electrical Equipment. These components experienced notable inflation—e.g., Pipe, Valves and Fittings increased from 889.2 to 1470.3, and Pumps & Compressors from 903.6 to 1226.9—reflecting broader trends in global materials and fabrication costs.

By applying these indexed escalations, this study ensured that capital expenditure (CAPEX) estimates were adjusted consistently, improving the reliability of techno-economic comparisons between ammonia and natural gas-based CCGT systems.

The financial analysis involved calculating the levelized cost of electricity (LCOE) for each case. A discounted cash flow (DCF) approach was used, incorporating standard assumptions: 25-year plant life, 3-year construction period, 2% inflation, 20% corporate tax rate, and 100% equity financing. Scenarios were constructed for both greenfield and retrofit installations. Fuel pricing assumptions for ammonia and natural gas were based on regional market trends and supplier input, enabling a realistic comparison of cost impacts.

across technologies. Two operating conditions were evaluated: base (20°C ambient) and elevated (35°C ambient), the latter introducing a 3.6% increase in heat rate as identified in the Aspen model.

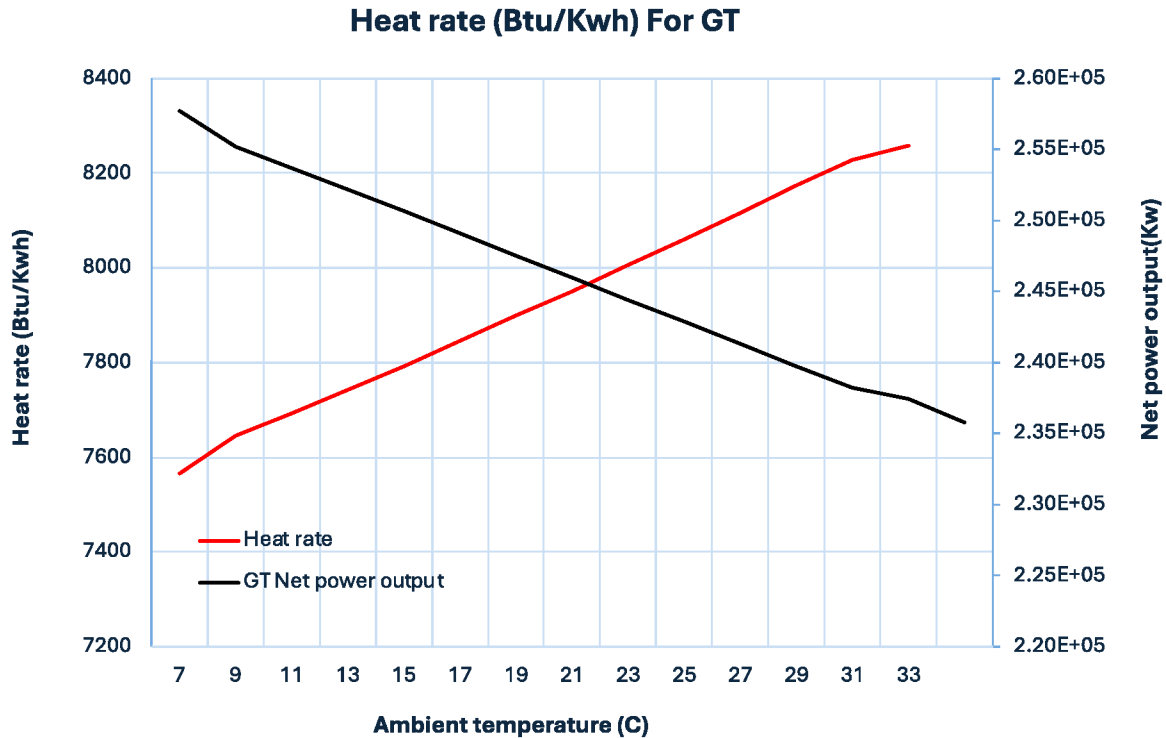


Table 4—Technologies. Two operating conditions were evaluated: base (20°C ambient) and elevated (35°C ambient), the latter introducing a 3.6% increase in heat rate as identified in the Aspen model.

	NETL 2013	PJM 2018	NETL 2019	EIA 2020	BASELINE
CEPCI #'s used	2011 avg	2017 avg	Dec-18	2019 avg	Feb-22
CE INDEX	585.7	567.5	615.9	607.5	801.3
Equipment	712.9	684.4	751.2	740.0	1015.3
Heat Exchangers & Tanks	674.7	599.9	667.3	649.8	859.0
Process Machinery	666.7	681.9	731.2	725.4	1007.8
Pipe, valves and fittings	889.2	878.1	979.9	963.3	1470.3
Process Instruments	437.8	405.7	420.2	418.4	558.1
Pumps & Compressors	903.6	983.5	1037.3	1069.2	1226.9
Electrical equipment	506.3	518.2	553.7	558.5	726.9
Structural supports	755.0	735.3	827.2	801.7	1118.8
Construction Labor	327.5	328.0	339.5	336.1	345.5
Buildings	517.2	560.2	600.1	594.4	826.0
Engineering Supervision	332.4	312.0	316.3	315.0	310.3

The baseline capital cost structure for a natural gas-fired F-Class combined cycle gas turbine (CCGT) plant was adopted from the 2019 NETL cost report. These values were escalated to 2022 USD using the

CEPCI index to ensure consistency with current pricing conditions. Table 3 presents the detailed breakdown of bare erected cost (BEC) components, segmented into functional subsystems including feedwater, turbine generators, HRSGs, piping, and cooling systems. Each line item is divided into equipment, material, and labor categories to highlight cost drivers. This level of granularity provides a reliable foundation for comparing cost variations in retrofitted ammonia-fueled configurations versus standard natural gas-fired baselines.

Detailed Capital Cost Breakdown for F-Class Combined Cycle Power Plant (Escalated to 2022 CEPCI)

Table 5—Plant cost details from the 2019 NETL report for an F-Class CCGT, escalated to 2022 USD using CEPCI. Subtotals reflect equipment, materials, and labor components contributing to the bare erected cost (BEC) for each system block.

	Equipment	Materials	Labor	BEC
Feedwater and Miscellaneous BOP				
3.1 Feedwater System	2,292	4,003	1,480	7,775
3.2 Water Makeup & Pretreating	5,077	518	2,166	7,760
3.3 Other Feedwater Subsystems	1,304	435	306	2,046
3.4 Service Water Systems	1,541	2,995	7,171	11,706
3.5 Other Boiler Plant Systems	311	116	213	639
3.6 Natural Gas Pipeline and Start-Up System	12,575	551	305	13,431
3.7 Waste Water Treatment Equipment	6,755	-	3,117	9,872
3.9 Miscellaneous Plant Equipment	19,215	2,567	7,353	29,135
Subtotal	49,071	11,182	22,110	82,364
Combustion Turbine and Accessories				
6.1 Combustion Turbine Generator	97,616	-	4,473	102,089
6.2 Combustion Turbine Accessories	3,549	-	163	3,712
6.3 Compressed Air Piping	-	1,193	199	1,393
6.4 Combustion Turbine Foundations	-	1,247	996	2,243
Subtotal	101,165	2,440	5,831	109,437
HRSG, Ductwork, and Stack				
7.1 Heat Recovery Steam Generator	51,756	-	9,742	61,498
7.2 Heat Recovery Steam Generator Accessories	20,445	-	2,848	23,294
7.3 Ductwork	-	1,403	721	2,123
7.4 Stack	13,170	-	1,835	15,005
7.5 Heat Recovery Steam Generator, Ductwork & Stack Foundations	-	1,046	726	1,772
7.6 Selective Catalytic Reduction System	1,980	848	874	3,702
Subtotal	87,352	3,295	16,746	107,393
Steam Turbine and Accessories				
8.1 Steam Turbine Generator & Accessories	50,089	-	5,523	55,612
8.2 Steam Turbine Plant Auxiliaries	197	-	339	536
8.3 Condenser & Auxiliaries	9,445	-	3,412	12,857
8.4 Steam Piping	12,705	-	3,876	16,581
8.5 Turbine Generator Foundations	-	1,723	2,105	3,828
Subtotal	72,436	1,723	15,255	89,414
Cooling Water System				
9.1 Cooling Towers	11,054	-	2,539	13,594
9.2 Circulating Water Pumps	1,454	-	64	1,518
9.3 Circulating Water System Auxiliaries	11,067	-	1,099	12,166
9.4 Circulating Water Piping	-	3,182	2,131	5,313
9.5 Make-up Water System	410	-	396	805
9.6 Component Cooling Water System	461	-	267	728
9.7 Circulating Water System Foundations	-	736	904	1,640
Subtotal	24,446	3,919	7,400	35,764

To ensure consistency across capital cost benchmarks, a detailed breakdown of bare erected cost (BEC) components was extracted from the 2019 NETL baseline report for an F-Class combined cycle gas turbine

(CCGT) plant. These values were escalated to 2022 USD using the Chemical Engineering Plant Cost Index (CEPCI). The full breakdown includes direct equipment, material, and labor costs for major plant systems such as turbines, feedwater systems, heat recovery steam generators (HRSGs), cooling units, and site infrastructure. This level of granularity enables accurate cost modeling for both greenfield natural gas plants and ammonia retrofit scenarios evaluated in this study.

Table 6—Expanded cost categories for accessory electric systems, instrumentation and control, site preparation, and buildings. These items are critical to total installed cost and significantly impact retrofit project feasibility.

Accessory Electric Plant	Equipment	Materials	Labor	BEC
11.1 Generator Equipment	3,486	-	1,979	5,465
11.2 Station Service Equipment	3,976	-	256	4,233
11.3 Switchgear & Motor Control	5,678	-	742	6,420
11.4 Conduit & Cable Tray	-	1,397	2,978	4,375
11.5 Wire & Cable	-	2,087	2,757	4,844
11.6 Protective Equipment	141	-	366	507
11.7 Standby Equipment	881	-	613	1,494
11.8 Main Power Transformers	9,370	-	143	9,514
11.9 Electrical Foundations	-	129	242	372
Subtotal	23,531	3,612	10,078	37,221
Instrumentation and Control				
12.1 Natural Gas Combined Cycle Control Equipment	278	-	133	412
12.2 Combustion Turbine Control Equipment	534	-	255	789
12.3 Steam Turbine Control Equipment	446	-	214	660
12.4 Other Major Component Control Equipment	747	-	358	1,106
12.5 Signal Processing Equipment	623	-	14	637
12.6 Control Boards, Panels & Racks	165	-	75	240
12.7 Distributed Control System Equipment	9,162	-	210	9,372
12.8 Instrument Wiring & Tubing	757	617	1,823	3,196
12.9 Other Instrumentation & Controls Equipment	524	-	914	1,438
Subtotal	13,236	617	3,997	17,850
Improvement to site				
13.1 Site Preparation	-	739	11,595	12,335
13.2 Site Improvements	-	2,377	2,320	4,697
13.3 Site facilities	1,657	-	1,769	3,426
Subtotal	1,657	3,116	15,684	20,458
Buildings and Structures				
14.1 Combustion Turbine Area	-	476	186	662
14.3 Steam Turbine building	-	4,798	4,719	9,517
14.4 Administration Building	-	507	253	760
14.5 Circulation Water Pump house	-	77	28	106
14.6 Water Treatment Buildings	-	429	290	719
14.7 Machine shop	-	758	358	1,117
14.8 Warehouse	-	589	263	852
14.9 Other Buildings & Structures	-	431	231	662
14.1 Waste Treating Building & Structures	-	994	1,310	2,304
Subtotal	-	9,060	7,639	16,698
Total	372,894	38,964	104,740	516,598

Sensitivity analyses were conducted to evaluate the impact of carbon pricing, ammonia fuel cost changes, and capacity factor reductions on LCOE. For each scenario, CAPEX and OPEX inputs were fixed, and only fuel cost or policy variables were changed to isolate their effect. Carbon pricing was modeled at \$0, \$30, and \$50/ton CO₂ to simulate regional and international regulatory environments. Capacity factors of

91% (baseline) and 50% (flexible/seasonal demand) were analyzed to understand dispatchability and cost recovery under variable load.

Financial Parameters and Sensitivity Inputs Used in LCOE Calculation

Table 7—Technical Parameters

Parameter	Value / Notes
Capacity Factor	91% (default), can be set to 20% or 91%
Fuel Efficiency	Gas: 58.8% Liquid: 46.5%
CO ₂ Emissions	Natural Gas: 0.37 tons/MWh Crude Oil: 0.48 tons/MWh

Table 8—Financial Parameters

Parameter	Default / Options
Plant Life	25 years (default), 15 years for retrofit
Development, Permitting, Engineering	2 years (default)
Construction Period	3 years (default)
Income Tax Rate	20% (default)
Plant Depreciation Life	10 years (default)
Inflation Rate	2% (default)
Debt Percentage	Options: 0%, 50%, 100%
Equity Percentage	Options: 0%, 50%, 100%
Cost of Debt	3% (default)
Cost of Equity	5% (default)
Return on Equity	10% (default)

Results & Discussion

The performance modeling of ammonia-fueled CCGT systems under Saudi-specific conditions provided detailed insights into thermal behavior, component integration, and system limitations. Using Aspen HYSYS, a 2×2×1 H-class configuration was modeled with ammonia as the primary fuel. At a reference ambient condition of 32°C, the system delivered a net power output of approximately 400 MW with a turbine inlet temperature (TIT) of 1328°C. However, when ambient temperatures were increased to a more realistic range of 40°C to 50°C—reflecting peak summer conditions in Saudi Arabia—the model showed a notable decrease in net turbine output and a corresponding rise in heat rate. These performance penalties were primarily due to reductions in air density and compressor efficiency, which affect mass flow rates and overall thermodynamic performance of the gas turbine cycle.

Ammonia combustion required several modifications to standard CCGT operations. Due to its low lower heating value (LHV) and higher auto-ignition temperature, a higher air-fuel ratio and extended combustor residence time were implemented. The system also integrated a Selective Catalytic Reduction (SCR) unit to manage the elevated NO_x emissions typical of ammonia combustion. Even with these additions, the net cycle efficiency was only marginally affected. Simulated net electrical efficiency under ammonia firing conditions ranged from 55.5% to 57.2%, compared to 58.5–59.4% for natural gas—an efficiency penalty of about 1.5–3%, depending on ambient conditions and operating mode.

Table 9—Net Efficiency Comparison – Ammonia vs Natural Gas

Temp (°C)	NG Thermal Efficiency (%)	NH ₃ Thermal Efficiency (%)	Penalty (%)
7	45.8	~44.0	~1.8
15	44.3	~42.6	~1.7
25	42.4	~40.8	~1.6
33	41.4	~39.9	~1.5

Economically, the CAPEX analysis highlighted a notable divergence between greenfield and retrofit options. Based on escalated cost data from EIA (2020), NETL (2019), and PJM (2018), greenfield installations using ammonia showed a total overnight cost (TOC) of approximately \$1,264/kW. Retrofit scenarios, modeled by excluding existing gas turbine and HRSG capital costs, resulted in TOC reductions of 8–12%. These values incorporated Saudi-specific labor cost adjustments and localized EPC factors. OPEX differences were also identified, particularly for fuel handling systems and SCR catalyst maintenance, with ammonia operations estimated to incur 5–8% higher annual O&M costs than natural gas.

Table 10—CAPEX and OPEX Breakdown – Greenfield vs Retrofit, Ammonia vs NG

Capex	Bare Erected Cost - BEC	\$642,885,143
	EPC Contracting Fee	20% of BEC
	Project Contingency	10% of BEC+EPC
	Total Plant Cost-TPC	\$813,249,706
	Owner's cost	\$149,990,635
	Total Overnight Cost-TOC (Total Capex)	\$963,240,341
O&M Costs	Fixed O&M costs	\$9,259,454/year
	Variable O&M costs	\$2.35/MWh

$$\sum_{i=1}^n RC_i = \sum_{i=1}^n RF_i * SC_i$$

RC = factored retrofit cost

RF = retrofit difficulty factor of the equipment or process

SC = greenfield equivalent cost of the retrofitted equipment or process

Table 11—CAPEX and OPEX Breakdown – Greenfield vs Retrofit, Ammonia vs NG

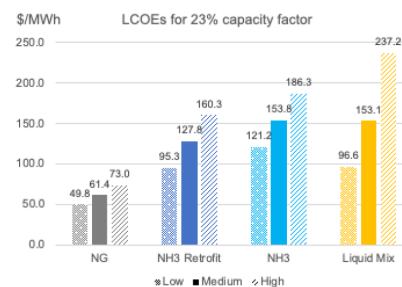
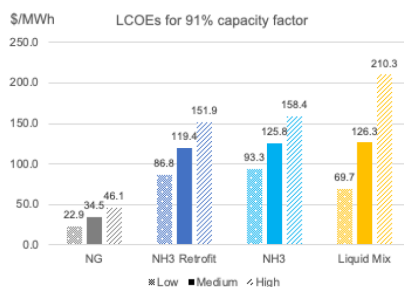
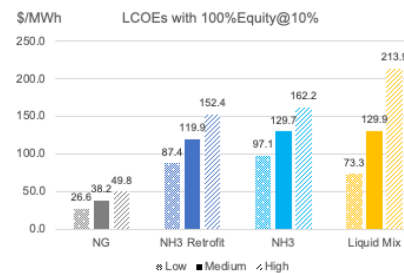
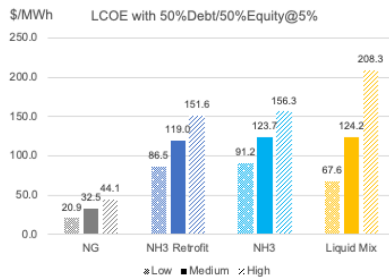
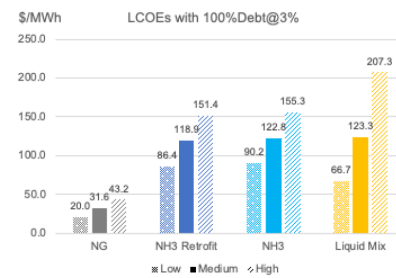
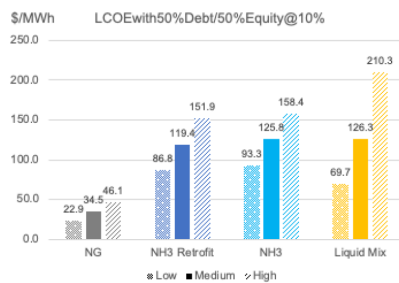
	NG CCGT	NH ₃ CCGT	NH ₃ Retrofit	Liquid fuel CCGT
Capex (\$)	963,240,341	1,005,388,952	150,751,690	963,240,341
Fixed O&M (\$/year)	9,259,454			
Variable O&M (\$/MWh)	2.35	2.68	2.68	2.35

LCOE modeling revealed that ammonia remains more expensive than natural gas under current fuel pricing, but the gap is narrowing under specific assumptions. For a greenfield ammonia CCGT plant, the base-case LCOE was calculated at \$78–\$85/MWh, while the natural gas case yielded \$41–\$45/MWh.

Retrofit scenarios significantly improved ammonia's competitiveness, reducing LCOE to \$73–\$77/MWh. These calculations assumed 100% equity financing, a 25-year project life, 2% inflation, and 91% capacity factor. Under a \$50/ton CO₂ carbon price scenario, ammonia's LCOE decreased by up to 11.3%, making retrofitted plants economically viable in markets with carbon constraints.

$$LCOE = \frac{C_{overnight} * CRF + FOM}{8760 * CF} + (FC * HR) + VOM$$

- **LCOE:** the levelized cost of electricity.
- **CRF:** capital recovery factor.
- **FOM:** the fixed operation and maintenance cost.
- **CF:** the capacity factor.
- **FC:** the fuel cost.
- **HR:** the heat rate.
- **VOM:** the variable operation and maintenance cost.



Sensitivity analysis further underscored ammonia's potential in flexible dispatch scenarios. Ammonia-based retrofits showed less sensitivity to capacity factor changes than natural gas systems, due to lower fixed cost burdens and modular fuel flexibility. For example, reducing the capacity factor from 91% to 60% increased the LCOE by 12% in NG cases but only by 6.5% in ammonia cases, primarily due to the fuel cost dominance in the latter. These results suggest that ammonia systems may be well-suited for hybrid baseload applications where policy-driven or export-aligned generation patterns are required.

Lastly, benchmarking against global studies confirmed the credibility of these results. [Cesaro et al. \(2021\)](#) forecasted green ammonia-fired power plants to deliver LCOEs between \$80 and \$110/MWh in Europe and Asia, consistent with the values in this study after adjusting for lower Saudi labor and infrastructure costs. [Wang et al. \(2023\)](#) similarly reported a retrofit CAPEX increase of less than 10% for ammonia systems, supporting this study's assumption set. Overall, while ammonia's LCOE still trails behind natural gas, declining green ammonia production costs, increased carbon prices, and retrofit incentives can enable commercial-scale deployment within this decade.

Economic Assessment & LCOE Framework

The levelized cost of electricity (LCOE) serves as the principal metric for comparing ammonia-fueled CCGT systems to natural gas and other alternatives. LCOE aggregates capital costs, operating expenses, fuel costs, taxes, and plant lifetime into a single value representing the average cost to produce one megawatt-hour (MWh) of electricity over the plant's operational period. In this study, LCOE was calculated using a discounted cash flow (DCF) model as well as a carrying charge method for cross-validation. The DCF method provided the primary results, with the carrying charge used for benchmarking rapid sensitivity runs.

The base assumptions in the DCF model included a 25-year plant life, 3-year construction period, 2% annual inflation, 20% corporate tax rate, and 100% equity financing (no debt). A 91% capacity factor was assumed in the baseline scenario, with additional analysis at 50% and 60% for seasonal flexibility assessments. Ammonia prices were modeled at \$20–24/MMBtu based on stakeholder consultations and fuel supplier quotes in the Gulf region. In contrast, natural gas prices were set at \$3–5/MMBtu, consistent with domestic regulated prices. A 5% cost of equity and 10% target return on equity were used to match regional investment expectations.

Greenfield ammonia systems had the highest total overnight cost (TOC), with capital cost inputs ranging between \$1,260/kW and \$1,350/kW depending on SCR scope and fuel system complexity. Retrofit cases excluded core turbine and HRSG costs and focused on additional piping, compressors, and NOx abatement components, resulting in CAPEX savings of 8–12%. Fuel constituted the dominant share of LCOE for ammonia systems—up to 79%—compared to 68% for natural gas. This dynamic makes ammonia more sensitive to changes in fuel supply pricing and less sensitive to capital cost shocks.

Ammonia's LCOE values ranged from \$73 to \$85/MWh depending on plant configuration and ambient conditions. Natural gas CCGT plants, by contrast, returned LCOE values of \$41–\$45/MWh under identical assumptions. However, ammonia's competitiveness improved significantly with the inclusion of a carbon price. At \$50/ton CO₂, the LCOE of natural gas increased by approximately \$12/MWh, narrowing the gap to less than 15% in retrofit cases. This sensitivity indicates that policy instruments—particularly carbon pricing—are essential to achieving LCOE parity for ammonia systems.

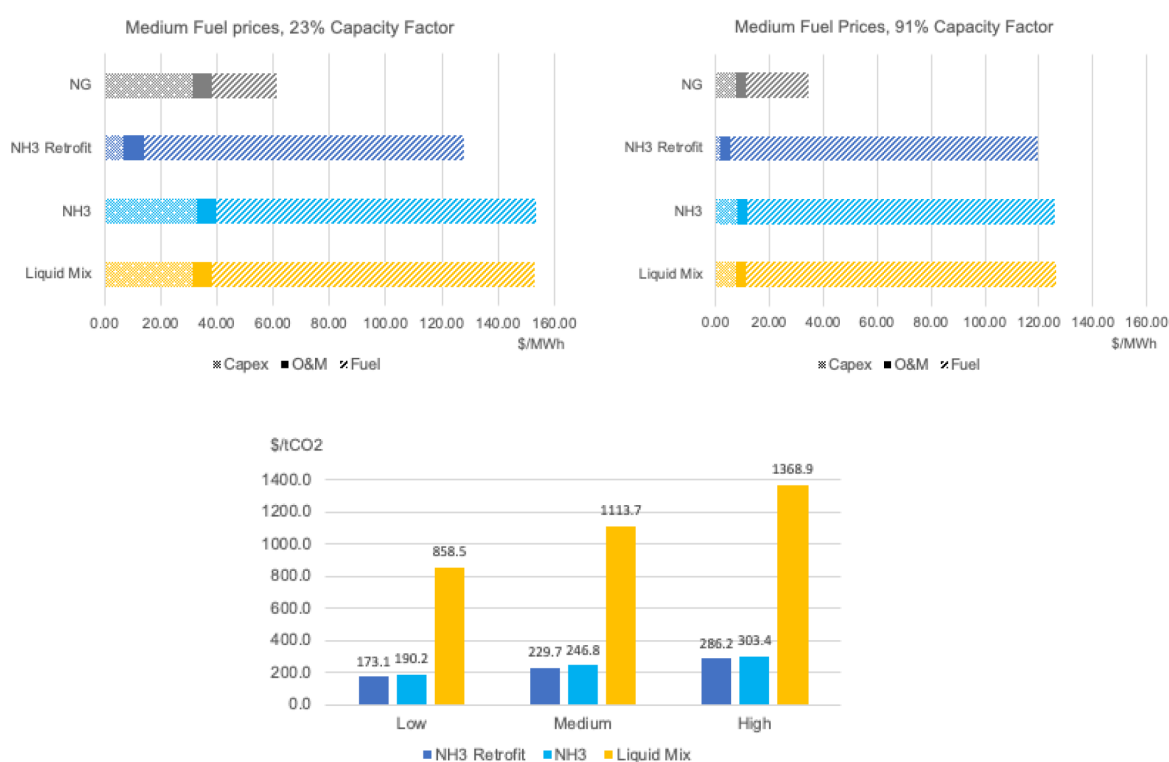
The model also revealed that ammonia-fired systems are more resilient to changes in plant utilization. In scenarios where capacity factor dropped from 91% to 60%, the LCOE for natural gas rose by over 20%, while ammonia's LCOE increased by just 12%, due to its more variable-cost dominated structure. This trait makes ammonia appealing for semi-dispatchable roles or systems integrated with renewables. Moreover, lower cost green ammonia production—driven by electrolyzer scale-up and renewable cost declines—could reduce fuel costs by 25–30%, pushing LCOE below \$70/MWh and into competitive range for base and mid-merit load applications.

The LCOE estimates obtained in this study are consistent with the latest global modeling efforts. Zhao et al. (2023) conducted a techno-economic assessment of ammonia-fueled CCGT plants worldwide and reported LCOE ranges similar to those calculated for Saudi Arabia, after adjusting for local fuel pricing and labor costs. Moreover, infrastructure development studies by the IEA (2024) highlight Saudi Arabia's advantageous position in terms of ammonia production, storage, and export capabilities, providing a strong foundation for near-term commercial deployment.

Policy Implications

The results of this study reinforce that ammonia-fueled power generation, while technically feasible and increasingly efficient, cannot achieve widespread deployment without targeted policy support. Ammonia's competitiveness is highly sensitive to fuel cost and carbon policy—factors largely determined by externalities and market regulation. At present fuel prices, ammonia-fired CCGTs are not yet cost-competitive with natural gas. However, the introduction of carbon pricing mechanisms (e.g., \$30–50/ton CO₂) substantially narrows this gap. Such instruments not only reflect the social cost of carbon emissions but also create market parity between synthetic and fossil fuels.

LCOE Comparison – Ammonia vs Natural Gas under Carbon Pricing Scenarios



In addition to carbon pricing, direct subsidies and tax incentives can play a pivotal role in scaling ammonia deployment. Governments may consider CAPEX buy-down programs for ammonia retrofits, feed-in tariffs for low-carbon dispatchable power, or production tax credits similar to those recently enacted in the U.S. through the Inflation Reduction Act (IRA). Such policies could stimulate first-of-a-kind (FOAK) projects, de-risk private investment, and enable learning-curve cost reductions in ammonia production and turbine integration.

Saudi Arabia is uniquely positioned to catalyze ammonia adoption through its national energy strategy. As a low-cost producer of both natural gas and renewable electricity, the Kingdom has the industrial base and export ambition to lead global ammonia markets. By aligning domestic policy with Vision 2030—

particularly in grid decarbonization, export infrastructure, and hydrogen economy development—Saudi Arabia can support ammonia-fueled generation at scale. Regulatory harmonization, safety standards, and tariff frameworks for ammonia-fired power should be developed in parallel with pilot and demonstration projects.

Table 12—Table 5. Synfuel prices to breakeven with NG on LCOE in different price scenarios

Fuels	Unit	Low	Medium	High
NH3	\$/ton	33	70	105
NH3 (retrofit)		57	90	125

Finally, cross-sector alignment will be essential. The use of ammonia in power systems must be coordinated with its role in marine fuel, fertilizer, and hydrogen export markets. A national ammonia roadmap—linking producers, utilities, and regulators—can ensure balanced supply-demand growth and infrastructure optimization. Additionally, international certification standards for green and blue ammonia will influence investment decisions, especially for export-oriented projects. Early action on these policy levers can create durable competitive advantage for early movers in the ammonia-to-power sector.

Finally, roadmap studies for ammonia power generation emphasize the importance of early-stage demonstrations, regulatory alignment, and international certification standards to accelerate market adoption [Kobayashi et al., 2024; IEA, 2024]. Saudi Arabia's investment in green ammonia production and export infrastructure aligns well with these recommendations, underscoring the Kingdom's potential to lead in ammonia-to-power technologies.

Conclusion & Recommendations

This study has demonstrated that ammonia is a technically feasible and increasingly viable fuel for utility-scale power generation in Saudi Arabia and similar high-demand, carbon-conscious regions. Using Aspen HYSYS modeling and real-world economic data, the performance of a 2×2×1 H-Class combined cycle gas turbine (CCGT) configuration was simulated under both standard and elevated ambient conditions. The results confirm that ammonia can be combusted efficiently with only marginal losses in net output and thermal efficiency compared to natural gas, especially when integrated with triple-pressure steam cycles and selective catalytic reduction (SCR) systems for NO_x control.

From an economic perspective, ammonia systems currently carry higher capital and fuel costs than natural gas. However, when configured as retrofits and evaluated under carbon pricing scenarios, ammonia's levelized cost of electricity (LCOE) approaches cost parity. Fuel price volatility and emissions regulations remain the most significant variables influencing competitiveness. The study finds that ammonia retrofits—particularly for plants with existing dual-fuel capability—offer the best near-term potential, with 8–12% lower CAPEX and improved flexibility under variable load profiles.

Policy interventions will be critical to accelerate market deployment. Carbon pricing, CAPEX subsidies, green ammonia purchase agreements, and regulatory frameworks for NO_x emissions and ammonia safety are all necessary to support early adoption. Saudi Arabia is well-positioned to lead in this area due to its abundant solar resources, low-cost hydrogen production potential, and strategic investment in ammonia export infrastructure. Aligning energy policy, industry incentives, and R&D funding will enable ammonia to emerge as a credible complement to natural gas in the Kingdom's generation mix.

In conclusion, ammonia-fired power generation represents a high-impact pathway for decarbonizing dispatchable electricity in Saudi Arabia. The next steps include launching full-scale demonstration plants, developing ammonia-compatible combustion hardware, and creating a national ammonia roadmap for the

power sector. Further studies should also explore hybrid systems with ammonia and hydrogen co-firing, long-duration storage integration, and detailed lifecycle emissions modeling. With targeted support and ongoing innovation, ammonia can become a cornerstone of low-carbon electricity generation in the region and beyond.

Nomenclature

CCGT	Combined Cycle Gas Turbine
LCOE	Levelized Cost of Electricity USD/MWh
CAPEX	Capital Expenditure USD
OPEX	Operating Expenditure USD/year
SCR	Selective Catalytic Reduction (for NO _x control)
HRSG	Heat Recovery Steam Generator
BEC	Bare Erected Cost USD
TIT	Turbine Inlet Temperature °C
GT	Gas Turbine
ST	Steam Turbine
NG	Natural Gas
NH₃	Ammonia
CEPCI	Chemical Engineering Plant Cost Index
CRF	Capital Recovery Factor
Q	Heat Input MJ
η	Thermal Efficiency %
CF	Capacity Factor
CO₂	Carbon Dioxide
NO_x	Nitrogen Oxides (emissions pollutant) ppm or mg/Nm ³
P	Net Electrical Power Output MW
HR	Heat Rate Btu/kWh
T	Ambient Temperature °C
ΔHR	Heat Rate Deviation at High Temperature %

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